

Yeji Kim

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EDUCATION

Stony Brook University

M.S. in Applied Mathematics and Statistics (Statistics Track) | GPA: 3.55

Stony Brook, New York

December 2024

Relevant Courses: Fundamentals of Computing | Design and Analysis of Categorical Data | Regression Theory | Statistical Learning | Big Data Analysis | Simulation and Modeling

Stony Brook University (SUNY Korea)

B.S. in Applied Mathematics and Statistics, magna cum laude | GPA: 3.75

Incheon, Republic of Korea

May 2023

Relevant Courses: Probability Theory | Data Mining | Operations Research II: Stochastic Models | Statistical Laboratory

SKILLS

Programming Languages: Python, R, and MATLAB

Data Analysis / ML: pandas, NumPy, scikit-learn, TensorFlow

Database: Oracle SQL, MySQL

Visualization: Tableau, ggplot2, Folium

Languages: Korean (Native), English (Advanced)

PROJECTS

Regression Model Comparison and Regularization

Aug 2023 – Dec 2023, Stony Brook, NY

- Implemented and evaluated OLS, Ridge, and Lasso regression models in Python
- Compared model performance using Mean Squared Error (MSE) and examined the effects of regularization on overfitting

Dimensionality Reduction and Visualization of the Iris Dataset Using PCA

Aug 2024 – Dec 2024, Stony Brook, NY

- Reduced the Iris dataset from 4 dimensions to 2 principal components while retaining over 95% of total variance
- Visualized PCA results for **3 Iris species** using R and ggplot2
- Analyzed clustering patterns and separation among the three species in the reduced feature space

Predicting Roadkill Risk in Hongcheon, Gangwon, South Korea

Jul 2025 – August 2025, Seoul, South Korea

- Built a machine learning pipeline to predict 3-level roadkill risk (0/1/2) in Hongcheon using road, wildlife-corridor, and environmental data
- Engineered spatial features including roadkill counts by road type and distance to the nearest wildlife corridor
- Compared SGD, ANN, and Gradient Boosting models; SGD achieved the highest validation Macro-F1 of 0.991 with 99.26% validation accuracy
- Visualized predicted roadkill risk using Folium and Tableau to highlight high-risk areas

Sunglasses Recommendation and Virtual Try-On

Sep 2025 - Oct 2025, Seoul, South Korea

- Built a face-shape classification and sunglasses recommendation system for five face-shape categories
- Trained and evaluated a CNN-based face-shape classification model using labeled image data
- Extracted facial landmarks with MediaPipe and implemented sunglasses positioning and overlay with OpenCV
- Developed a Streamlit web demo integrating image upload, face-shape prediction, sunglasses recommendation, and virtual try-on

WORK EXPERIENCE

SUNY KOREA

Incheon, Republic of Korea

Teaching Assistant

August 2019 – June 2021

- Held two hours of weekly office hours, supporting over 40 students in Precalculus, Linear Algebra, and Calculus II
- Led Precalculus recitation sessions for 40 undergraduate students
- Graded assignments and proctored exams across Precalculus, Linear Algebra, and Calculus II

Tutor

February 2020 – June 2020

- Tutored four students weekly in Calculus I, providing individualized support with course concepts and assignments